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Labsheet-2

1.Student Registration Form: Create a Student Registration Form using HTML. Include text fields, radio buttons, checkboxes, dropdown list, textarea, submit, and reset buttons. Apply CSS to style the form.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }

    .container {
      width: 500px;
      margin: 30px auto;
      background-color: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    h1 {
      text-align: center;
```

```
    color: #333;
}

label {
    font-weight: bold;
}

input[type="text"],
input[type="email"],
input[type="password"],
input[type="date"],
select,
textarea {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    margin-bottom: 15px;
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
}

.gender,
.hobbies {
    margin-bottom: 15px;
}
```

```
    input[type="submit"],
    input[type="reset"] {
      padding: 10px 25px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      margin-right: 10px;
    }

    input[type="submit"] {
      background-color: green;
      color: white;
    }

    input[type="reset"] {
      background-color: red;
      color: white;
    }

    input[type="submit"]:hover,
    input[type="reset"]:hover {
      opacity: 0.8;
    }
  </style>
</head>
```

```
<body>

  <div class="container">

    <h1>Student Registration Form</h1>

    <form>

      <!-- Text Field -->
      <label>Name:</label>
      <input type="text" name="name" placeholder="Enter your name">

      <!-- Email Field -->
      <label>Email:</label>
      <input type="email" name="email" placeholder="Enter your email">

      <!-- Password Field -->
      <label>Password:</label>
      <input type="password" name="password" placeholder="Enter password">

      <!-- Radio Buttons -->
      <label>Gender:</label>
      <div class="gender">
        <input type="radio" name="gender" value="male"> Male
        <input type="radio" name="gender" value="female"> Female
```

```
    <input type="radio" name="gender" value="other"> Other
</div>

<!-- Date Field -->
<label>Date of Birth:</label>
<input type="date" name="dob">

<!-- Dropdown List -->
<label>Course:</label>
<select name="course">
    <option value="">-- Select Course --</option>
    <option value="bca">BCA</option>
    <option value="bba">BBA</option>
    <option value="bcom">B.Com</option>
    <option value="btech">B.Tech</option>
</select>

<!-- Checkboxes -->
<label>Hobbies:</label>
<div class="hobbies">
    <input type="checkbox" name="hobby" value="reading"> Reading
    <input type="checkbox" name="hobby" value="coding"> Coding
    <input type="checkbox" name="hobby" value="sports"> Sports
    <input type="checkbox" name="hobby" value="music"> Music
</div>
```

```
        <!-- Textarea -->
        <label>Address:</label>
        <textarea name="address" rows="5"
              placeholder="Enter your address"></textarea>

        <!-- Submit and Reset Buttons -->
        <input type="submit" value="Submit">
        <input type="reset" value="Reset">

    </form>

  </div>

</body>
</html>
```

2.College Website using Frames: Create a college website using HTML Frames with Header, Navigation Menu, and Main Content sections. Add navigation links that display different pages in the main frame.

```
<!DOCTYPE html>
<html>
<head>
  <title>College Website</title>
</head>

<frameset rows="20%,80%">

  <!-- Header and Main Section -->
```

```
  <frameset cols="100%">
    <frame src="header.html" name="header">
  </frameset>

  <frameset cols="25%,75%">
    <!-- Navigation Menu -->
    <frame src="menu.html" name="menu">

    <!-- Main Content -->
    <frame src="home.html" name="main">
  </frameset>

</frameset>

</html>
<!DOCTYPE html>
<html>
<head>
  <title>Header</title>
</head>
<body style="background-color: lightblue; text-align: center;">

  <h1>COER University</h1>
  <h3>Welcome to Our College Website</h3>

</body>
```

```
</html>
<!DOCTYPE html>
<html>
<head>
  <title>Navigation Menu</title>
</head>
<body>

  <h2>Navigation Menu</h2>

  <ul>
    <li>
      <a href="home.html" target="main">Home</a>
    </li>

    <li>
      <a href="about.html" target="main">About Us</a>
    </li>

    <li>
      <a href="courses.html" target="main">Courses</a>
    </li>

    <li>
      <a href="contact.html" target="main">Contact Us</a>
    </li>
```

```
  </ul>

</body>
</html>
<!DOCTYPE html>
<html>
<head>
  <title>Home</title>
</head>
<body>

  <h1>Home</h1>

  <p>
    Welcome to COER University. Our college provides quality education
    and a supportive learning environment for students.
  </p>

</body>
</html>
<!DOCTYPE html>
<html>
<head>
  <title>About Us</title>
</head>
<body>
```

```
  <h1>About Us</h1>

  <p>
    COER University is committed to providing quality education,
    practical knowledge, and opportunities for students to develop
    their skills.
  </p>

</body>
</html>
<!DOCTYPE html>
<html>
<head>
  <title>Courses</title>
</head>
<body>

  <h1>Courses Offered</h1>

  <ul>
    <li>Bachelor of Computer Applications (BCA)</li>
    <li>Bachelor of Business Administration (BBA)</li>
    <li>Bachelor of Commerce (B.Com)</li>
    <li>Bachelor of Technology (B.Tech)</li>
  </ul>
```

```
</body>
</html>
<!DOCTYPE html>
<html>
<head>
  <title>Contact Us</title>
</head>
<body>

  <h1>Contact Us</h1>

  <p>
    <b>College:</b> COER University
  </p>

  <p>
    <b>Email:</b> info@college.com
  </p>

  <p>
    <b>Phone:</b> 9876543210
  </p>

</body>
</html>
```

3.Login and Admission Form: Design a Login Form and College Admission Form using HTML. Use appropriate form elements and apply CSS for formatting, borders, spacing, fonts, and buttons.

```
<!DOCTYPE html>
<html>
<head>
  <title>Login and College Admission Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
      padding: 20px;
    }

    h1 {
      text-align: center;
      color: #333;
    }

    .container {
      display: flex;
      justify-content: center;
      gap: 30px;
      flex-wrap: wrap;
    }
```

```
.form-box {
  width: 400px;
  background-color: white;
  padding: 25px;
  border: 2px solid #ccc;
  border-radius: 10px;
  box-shadow: 0 0 10px gray;
}

h2 {
  text-align: center;
  color: #444;
}

label {
  font-weight: bold;
  display: block;
  margin-top: 12px;
  margin-bottom: 5px;
}

input[type="text"],
input[type="email"],
input[type="password"],
input[type="date"],
```

```
select,
textarea {
  width: 100%;
  padding: 10px;
  border: 1px solid #aaa;
  border-radius: 5px;
  box-sizing: border-box;
  font-size: 14px;
}

textarea {
  resize: vertical;
}

.radio-group,
.checkbox-group {
  margin: 8px 0;
}

.radio-group label,
.checkbox-group label {
  display: inline;
  font-weight: normal;
  margin-right: 10px;
}
```

```
.buttons {
   text-align: center;
   margin-top: 20px;
}

input[type="submit"],
input[type="reset"] {
   padding: 10px 25px;
   border: none;
   border-radius: 5px;
   margin: 5px;
   font-size: 15px;
   cursor: pointer;
}

input[type="submit"] {
   background-color: green;
   color: white;
}

input[type="reset"] {
   background-color: red;
   color: white;
}

input[type="submit"]:hover,
```

```
    input[type="reset"]:hover {
        opacity: 0.8;
    }
  </style>
</head>

<body>

  <h1>College Portal</h1>

  <div class="container">

    <!-- Login Form -->
    <div class="form-box">

      <h2>Login Form</h2>

      <form>

        <label for="username">Username:</label>
        <input type="text" id="username"
            placeholder="Enter username">

        <label for="login-email">Email:</label>
        <input type="email" id="login-email"
            placeholder="Enter email">
```

```
      <label for="password">Password:</label>
      <input type="password" id="password"
          placeholder="Enter password">

      <div class="buttons">
        <input type="submit" value="Login">
        <input type="reset" value="Reset">
      </div>

    </form>

</div>



<!-- College Admission Form -->
<div class="form-box">

    <h2>College Admission Form</h2>

    <form>

      <label for="name">Student Name:</label>
      <input type="text" id="name"
          placeholder="Enter your name">
```

```
<label for="email">Email:</label>
<input type="email" id="email"
     placeholder="Enter your email">

<label for="dob">Date of Birth:</label>
<input type="date" id="dob">

<label>Gender:</label>

<div class="radio-group">
   <input type="radio" name="gender" value="male">
   <label>Male</label>

   <input type="radio" name="gender" value="female">
   <label>Female</label>

   <input type="radio" name="gender" value="other">
   <label>Other</label>
</div>

<label for="course">Select Course:</label>

<select id="course">
   <option value="">-- Select Course --</option>
   <option value="bca">BCA</option>
   <option value="bba">BBA</option>
```

```
    <option value="bcom">B.Com</option>
    <option value="btech">B.Tech</option>
</select>

<label>Hobbies:</label>

<div class="checkbox-group">
    <input type="checkbox" name="hobby" value="reading">
    <label>Reading</label>

    <input type="checkbox" name="hobby" value="coding">
    <label>Coding</label>

    <input type="checkbox" name="hobby" value="sports">
    <label>Sports</label>
</div>

<label for="address">Address:</label>
<textarea id="address" rows="4"
        placeholder="Enter your address"></textarea>

<div class="buttons">
    <input type="submit" value="Submit">
    <input type="reset" value="Reset">
</div>
```

```
        </form>

    </div>

  </div>

</body>

</html>
```

4.JavaScript Calculator: Create a simple calculator using HTML and JavaScript to perform addition, subtraction, multiplication, and division of two numbers.

```
<!DOCTYPE html>

<html>

<head>

  <title>Simple Calculator</title>


  <style>

    body {

      font-family: Arial, sans-serif;

      background-color: #f2f2f2;

      text-align: center;

      padding: 50px;

    }


    .calculator {

      width: 350px;

      margin: auto;
```

```
        padding: 25px;
        background-color: white;
        border: 2px solid #333;
        border-radius: 10px;
    }

    input {
        width: 90%;
        padding: 10px;
        margin: 10px;
        font-size: 16px;
    }

    button {
        padding: 10px 15px;
        margin: 5px;
        font-size: 16px;
        cursor: pointer;
    }

    #result {
        font-size: 20px;
        font-weight: bold;
        margin-top: 20px;
    }
</style>
```

```
</head>

<body>

  <div class="calculator">

    <h1>Simple Calculator</h1>

    <input type="number" id="num1" placeholder="Enter first number">

    <input type="number" id="num2" placeholder="Enter second number">

    <br>

    <button onclick="add()">+</button>
    <button onclick="subtract()">−</button>
    <button onclick="multiply()">×</button>
    <button onclick="divide()">÷</button>

    <p id="result">Result: </p>

  </div>

  <script>

    function getNumbers() {
```

```
    let a = parseFloat(document.getElementById("num1").value);
    let b = parseFloat(document.getElementById("num2").value);

    return [a, b];
}

function add() {
    let [a, b] = getNumbers();
    document.getElementById("result").innerHTML =
        "Result: " + (a + b);
}

function subtract() {
    let [a, b] = getNumbers();
    document.getElementById("result").innerHTML =
        "Result: " + (a - b);
}

function multiply() {
    let [a, b] = getNumbers();
    document.getElementById("result").innerHTML =
        "Result: " + (a * b);
}

function divide() {
    let [a, b] = getNumbers();
```

```
        if (b == 0) {
            document.getElementById("result").innerHTML =
                "Cannot divide by zero!";
        } else {
            document.getElementById("result").innerHTML =
                "Result: " + (a / b);
        }
    }

  </script>

</body>
</html>
```

5.JavaScript Validation: Create a Student Registration Form using HTML and JavaScript. Validate fields such as Name, Email, Mobile Number, Password, and Age before submitting the form. Display appropriate validation messages.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }
```

```
.container {
  width: 450px;
  margin: 40px auto;
  padding: 25px;
  background-color: white;
  border: 2px solid #333;
  border-radius: 10px;
}

h1 {
  text-align: center;
}

label {
  font-weight: bold;
}

input {
  width: 100%;
  padding: 10px;
  margin: 8px 0;
  box-sizing: border-box;
}

.error {
```

```
        color: red;
        font-size: 14px;
      }

      input[type="submit"],
      input[type="reset"] {
        width: 48%;
        background-color: green;
        color: white;
        border: none;
        cursor: pointer;
      }

      input[type="reset"] {
        background-color: red;
      }
    </style>
</head>

<body>

<div class="container">

    <h1>Student Registration</h1>

    <form onsubmit="return validateForm()">
```

```
<label>Name:</label>
<input type="text" id="name">
<div id="nameError" class="error"></div>

<label>Email:</label>
<input type="text" id="email">
<div id="emailError" class="error"></div>

<label>Mobile Number:</label>
<input type="text" id="mobile">
<div id="mobileError" class="error"></div>

<label>Password:</label>
<input type="password" id="password">
<div id="passwordError" class="error"></div>

<label>Age:</label>
<input type="number" id="age">
<div id="ageError" class="error"></div>

<br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">
```

```
  </form>

</div>

<script>

function validateForm() {

  let name = document.getElementById("name").value.trim();
  let email = document.getElementById("email").value.trim();
  let mobile = document.getElementById("mobile").value.trim();
  let password = document.getElementById("password").value;
  let age = document.getElementById("age").value;

  let valid = true;

  // Clear previous messages
  document.getElementById("nameError").innerHTML = "";
  document.getElementById("emailError").innerHTML = "";
  document.getElementById("mobileError").innerHTML = "";
  document.getElementById("passwordError").innerHTML = "";
  document.getElementById("ageError").innerHTML = "";

  // Name validation
  if (name == "") {
    document.getElementById("nameError").innerHTML =
```

```
        "Name is required.";
    valid = false;
} else if (!/^[A-Za-z ]+$/.test(name)) {
    document.getElementById("nameError").innerHTML =
        "Name should contain only letters.";
    valid = false;
}

// Email validation
let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

if (email == "") {
    document.getElementById("emailError").innerHTML =
        "Email is required.";
    valid = false;
} else if (!emailPattern.test(email)) {
    document.getElementById("emailError").innerHTML =
        "Enter a valid email address.";
    valid = false;
}

// Mobile number validation
if (mobile == "") {
    document.getElementById("mobileError").innerHTML =
        "Mobile number is required.";
    valid = false;
```

```
} else if (!/^[0-9]{10}$/.test(mobile)) {
   document.getElementById("mobileError").innerHTML =
      "Mobile number must contain exactly 10 digits.";
   valid = false;
}

// Password validation
if (password == "") {
   document.getElementById("passwordError").innerHTML =
      "Password is required.";
   valid = false;
} else if (password.length < 6) {
   document.getElementById("passwordError").innerHTML =
      "Password must be at least 6 characters.";
   valid = false;
}

// Age validation
if (age == "") {
   document.getElementById("ageError").innerHTML =
      "Age is required.";
   valid = false;
} else if (age < 18 || age > 60) {
   document.getElementById("ageError").innerHTML =
      "Age must be between 18 and 60.";
   valid = false;
```

```
    }

    if (valid) {
        alert("Registration successful!");
    }

    return valid;
}

</script>

</body>
</html>
```